Bivariate Data

Part A: Bivariate Data & Correlation

2unit

selective

Q 1 Sydney Tech

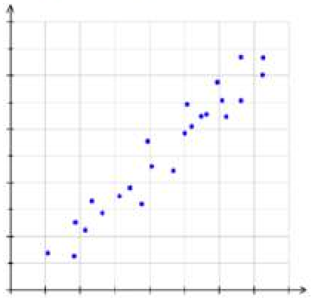
What is the correlation between the variables in this scatterplot?

(A) Weak positive

(B) Weak negative

(C) Moderate positive

(D) Moderate negative



Q 2 St George Girls

A scatterplot of pain (as reported by patients) compared to the dosage (in mg) of a drug is shown below.

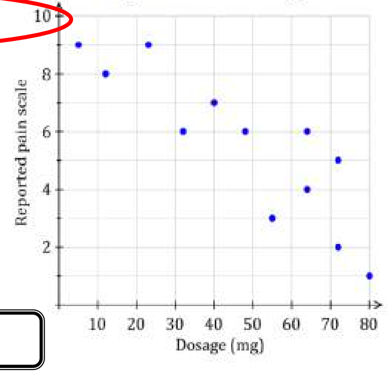
How could you describe the correlation between the pain and the dosage?

(A) A moderate negative correlation

(B) A moderate positive correlation

(C) A weak positive correlation.

(D) No correlation.



Q 3 St George Girls

A scatterplot is shown

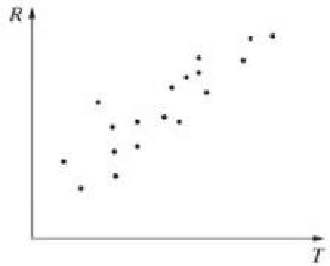
Which of the following best describes the correlation between R and T?

(A) Positive

(B) Negative

(C) Positively skewed

(D) Negatively skewed



Q 4 Hunters Hill

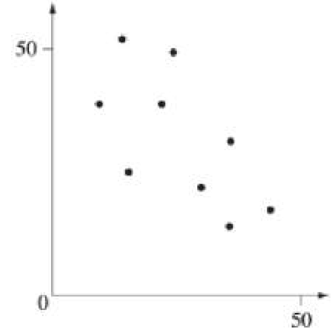
The graph shows a scatter plot for a set of data. Which of the following values is the most appropriate Pearson's correlation coefficient of this set of data?

A. -1

B. 0.35

C. -0.35

D. -0.63



Q 5 Western

Holly drew a scatter-plot of a binomial data set which compared the construction time of houses with their cost.

The construction times ranged from 6 weeks to 6 months.

She found the equation of the line of best fit and used it to estimate the cost of a house which took 10 months to build.

What term would describe this process?

A. Causality

B. Correlation

C. Extrapolation

D. Interpolation

Predicting values outside the original data set is called extrapolation.

Q 6 Ace

The Pearson's correlation coefficient between students assessment result and their height was 0.12. What is the meaning of this correlation? **2**

Assessment results increase as height increases.

Low positive correlation.

Not a strong relationship.

Bivariate Data

Part B: Least Squares Regression Line

2unit

selective

Q 1 Clancy

Paolo and Gianluca are watching Italy in the UEFA Euro 2020 Championship Final. It is such an exciting match that Paolo is worried that Gianluca will faint with excitement. He takes his pulse at various intervals during the game.
The information is recorded in the table and graphed on the scatterplot below.

t = time since the match started (mins)	2	5	8	16	20	30	42	48	53
b = heart rate (beats / min)	72	80	68	72	80	84	77	90	99

(a) Find the value of r, the correlation coefficient, correct to 2 decimal places. **1**

(b) Find the equation of the least-squares regression line that can be used to predict b from t. Write the gradient and y–intercept correct to 2 decimal places, using the variables given in the table. **2**

(c) What is Gianluca's predicted heart rate 46 minutes into the match? **1**

(d) The match ended after 150 minutes. **1**
Paolo approximated Gianluca's heart rate to be 130 beats / min at the conclusion of the match.
Is this a valid conclusion? Explain your answer.

a) Using calculator $r = 0.79672622281$
$r = 0.80$ (2 d.p.)

b) Using calculator y-int $= 70.17501318 = 70.18$
Gradient $= 0.4036825062 = 0.40$

$\therefore y = mx + c$
$b = mt + c$
$b = 0.40t + 70.18$

c) when $t = 46$ $b = 0.40(46) + 70.18$
$b = 88.58$ beats/min

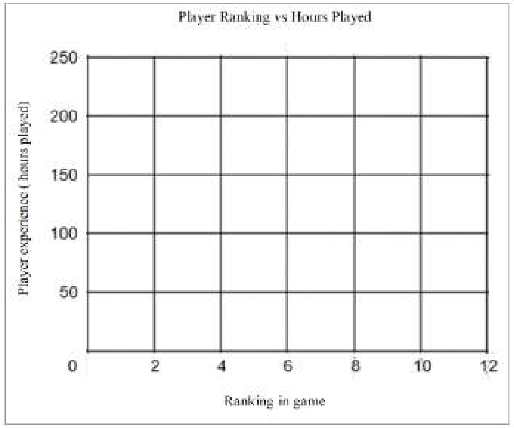
d) time (t) since the match started only ranges from 2 mins to 53 mins.
150 mins is outside this range.
We should not extrapolate beyond the given data set.
(It is not reasonable for his heartrate to be that high at 130 bpm at the end of the match)

Q 2 Hurlstone

Ten students were ranked on their computer gaming ability on a new game.
Each student also calculated the number of hours that they have played the game.
The results are recorded in the table below.

Rank	1	2	3	4	5	6	7	8	9	10
Hours Played	198	143	88	102	82	94	54	36	20	12

(i) Using the axes below draw the scatterplot for the data in the table. **1**



(ii) Calculate, to 2 decimal places, Pearson's correlation coefficient, r, and describe the relationship between a player's rank and the number of hours that they have played the game. **2**

(iii) Find the equation of the least-squares regression line for the data given above. Give your answer correct to 2 decimal places. **2**

(i)

(ii) $r = -0.94$
There is a very strong negative relationship between rank and gaming hours. As time played increases, rank decreases.

(iii) $a = 180.47$ $b = -17.74$
$y = -17.74x + 180.47$

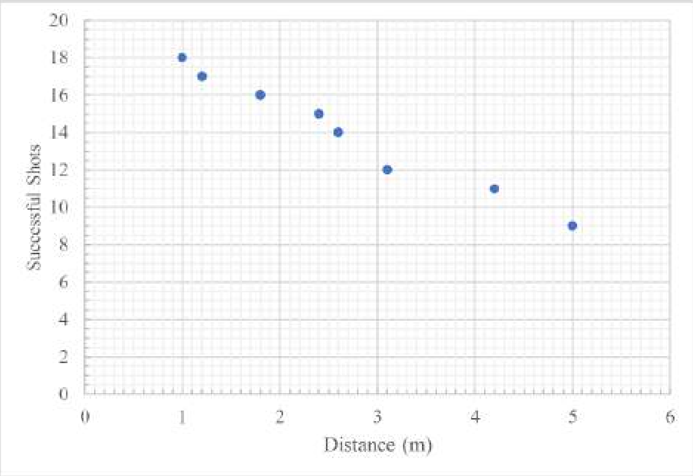
Bivariate Data

Part B: Least Squares Regression Line

Q 3 — Cerdon

A netball player took 20 shots from each of 8 different positions that were different distances from the goal. The results are shown in the table below.

Distance (m)	1.2	2.4	5.0	4.2	1.0	1.8	3.1	2.6
Successful Shots	17	15	9	11	18	16	12	14



(a) Using the scatterplot, describe the correlation between the two variables. (1 mark)

(b) Calculate Pearson's correlation coefficient for this data, correct to 3 decimal places. (1 mark)

(c) Determine the equation of the least-squares regression line for this data, rounding any values to 2 decimal places. (1 mark)

(d) Using your equation from part (c), predict the distance, to the nearest cm, at which the netball player will make 10 successful shots. (2 marks)

a). Very strong negative linear correlation

b). $r = -0.988$

c). $y = 19.85 - 2.198x$

Successful shots $= 19.85 - 2.198 \times$ Distance

d). $10 = 19.85 - 2.198 \times$ Distance

Distance $= \dfrac{10 - 19.85}{-2.198} = \boxed{4.48 \text{ m}}$

Q 4 — Gosford

Luke suspects that the rate at which he spends cash is affected by the amount of cash he withdrew at his previous visit to an ATM.

The table below shows the amount of cash withdrawn, $\$x$, from an ATM, and the time, y hours, until Luke's next withdrawal from an ATM.

Withdrawal	1	2	3	4	5	6	7	8	9	10
x	40	10	100	110	120	150	20	90	80	130
y	56	62	195	330	94	270	48	196	214	286

(a) Find the equation of the least squares regression line for y in terms of x, for the withdrawals 1 to 10 and hence estimate how much cash (to the nearest \$10) Luke would need to withdraw from the ATM at his previous visit in order to not need to visit an ATM again for 120 hours. **2**

(b) Calculate the correlation coefficient between x and y for the withdrawals 1 to 10. Describe the nature of the correlation. **2**

(a) from calculator: $y = A + Bx$

$A = 30.26$ $\quad y = 1.7x + 30.26$

$B = 1.7$

Find x at $y = 120$

$\therefore 120 = 1.7x + 30.26$

$\therefore x = 52.79$

$\therefore$ withdrawal $= \$60$

(b) $r = 0.777$ {strong} positive correlation

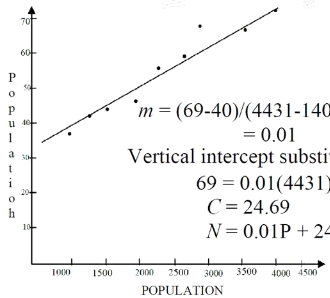
Q 5 — Gosford

The population of towns in Europe and the number of cars stolen is recorded below.

Population (P)	1159	1400	1654	2010	2453	2635	3120	3740	4431
Number of cars stolen (N)	35	40	43	46	55	57	66	63	69

(a) Graph the data on a scatterplot and draw a line of best fit to represent all data points. **2**

(b) Find the equation of the line of best fit. **1**



$m = (69-40)/(4431-1400)$
$= 0.01$

Vertical intercept substitute (4431,69)

$69 = 0.01(4431) + c$

$C = 24.69$

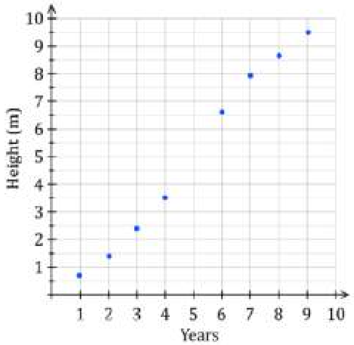
$N = 0.01P + 24.69$

Q 6 Sydney Tech

Tim is a scientist studying the growth of a particular tree over several years. The data he recorded is shown in the table below.

Years since planting, t	1	2	3	4	6	7	8	9
Height of tree, H metres	0.7	1.4	2.4	3.5	6.6	7.9	8.7	9.5

A scatterplot of the data is shown below.



(a) What is Pearson's correlation coefficient? Answer correct to 4 decimal places. 1

(b) Find the equation of the least-squares line of best fit in terms of years (t) and height (h). Answer correct to 2 decimal places. 1

(c) Tim did not record the tree's height after five years. Predict the height after five years, correct to 1 decimal place. 1

(d) Estimate how many years it will take for the tree to reach a height of 24 metres. Answer correct to 1 decimal place. 1

(a) $r = 0.9952$

(b) $y = Bx + A \quad A = -0.85$

$B = 1.19$

$\therefore h = 1.19t - 0.85$

(c) when $t = 5$, $h = 1.19(5) - 0.85 \quad = 5.1\text{m}$

(d) when $h = 24$, $24 = 1.19t - 0.85$

$24.85 = 1.19t$

$t = 20.9$ years

Q 7 Sydney Girls

The data in the following table relates study time and test scores. 2

Study time in hours	Test grade in percent
7	83
8	85
9	88
10	91
11	93
12	95

Find a regression line for the data above, using accuracy to two decimal places.

$y = bx + a$

$a \doteqdot 65.55$ (2dp) $\quad b \doteqdot 2.49$ (2dp)

Regression line: $\therefore y = 2.49x + 65.55$

Q 8 Manly

The body weights and metabolic rate recordings for five women are shown below.

Weight (kg)	52	63	65	47	49
Metabolic rate	1671	1669	1812	1442	1607

i. Find the correlation coefficient, r. (correct to 3 decimal places) 1

ii. Find the coefficients of the linear regression line, $y = mx + c$, where x is the weight and y is the metabolic rate. 1

iii. Verify that $m = r\dfrac{s_y}{s_x}$, where s_x and s_y are the standard deviations of the x and y values respectively. 2

i $\quad 0.8233 = \mathbf{0.823}$

ii $\quad m = 13.346$

$c = 903.499$

iii $\quad \left[\begin{array}{l} s_y = 119.732 \\ s_x = 7.386 \end{array}\right. \Rightarrow r \times \dfrac{s_y}{s_x} = 0.8233 \times 16.2096 = 13.346 = m$

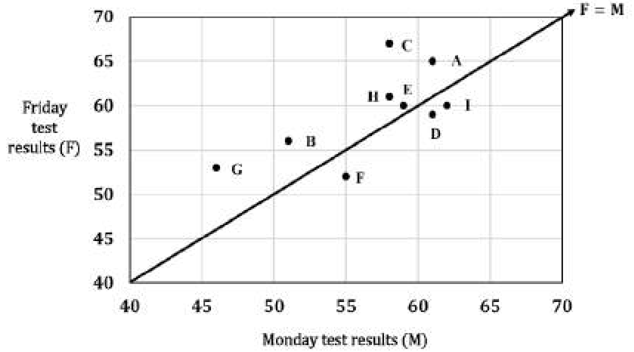
Bivariate Data

Part B: Least Squares Regression Line

Q 9 — St George Girls

After a class of ten students sits a test on Monday, the teacher spends time doing revision and tests the students again on Friday. Only nine of the students are present for the second test on Friday. The results, in percentages, for those nine students, are shown in the following table and scatterplot.

Student	A	B	C	D	E	F	G	H	I	J
Monday (M)	61	51	58	61	59	55	46	58	62	63
Friday (F)	65	56	67	59	60	52	53	61	60	Absent



(a) Using your calculator, find the correlation coefficient for the 9 students and explain the type and strength of correlation this data represents. **2**

(b) Determine the equation of the least squares-regression line for this data. **1**

(c) Using your answer to part (b), predict the result for the student who was absent for Friday's test. Give your answer to the nearest whole mark. **1**

(d) The scatterplot above also shows the line $F = M$. Explain the significance of a student being represented below the line $F = M$. **1**

(a) $r = 0.66020159\ldots \doteq 0.66$ (2 d.p.)
This is a moderate positive correlation

(b) $A = 23.80666\ldots$ $\quad B = 0.6237574\ldots$
$\doteq 23.81$ (2dp) $\quad \doteq 0.62$ (2dp)
$\therefore F = 23.81 + 0.62M$

(c) $F = 23.81 + 0.62(63)$
$= 62.87 \doteq 63$

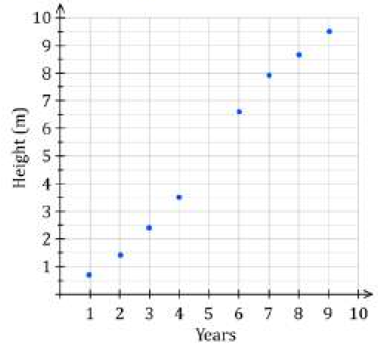
(d) Students below the line did better on Monday than Friday.

Q10 — Grafton

Hayden is an agricultural scientist studying the growth of a particular tree over several years. The data he recorded is shown in the table below.

Years since planting, t	1	2	3	4	5	6	7	8	9
Height of tree, H metres	0.7	1.4	2.4	3.5		6.6	7.9	8.7	9.5

A scatterplot of the data is shown below.



(a) What is Pearson's correlation coefficient? Answer correct to 4 decimal places. **1**

(b) Find the equation of the least-squares line of best fit in terms of years (t) and height (h). Answer correct to 2 decimal places. **2**

(c) Hayden did not record the tree's height after five years. Predict the height after five years, correct to one decimal place. **1**

(d) Use algebra to estimate how many years it will take for the tree to reach a height of 20 metres. Answer correct to 1 decimal place. **1**

(e) Comment on the reliability of your answers in (c) and (d). **1**

(a) $r = 0.995193611 \doteq 0.9952$ (calculator)

(b) $y = mx + b$
$h = BX + A$ $\quad \therefore h = 1.19t - 0.85$ (calculator)

(c) when $t = 5$: $h = 1.19 \times 5 - 0.85 \doteq 5.1$ m

(d) $h = 20$ find t $\quad 20 = 1.19t - 0.85$
$1.19t = 20.85 \quad | \div 1.19$
$t \doteq 17.5$ years

(e) strong positive association
Q33c) involves interpolation (very reliable)
Q33d) involves extrapolation (less reliable)

Q11 Baulkham Hills

The following information shows a group of people's waist measurements and weights.

Waist (cm) x	72	67	85	96	80	90	98	105
Weight (kg) y	58	50	72	85	70	79	82	84

(i) Calculate the correlation coefficient, r, for their waist and weight measurements correct to 3 decimal places and hence describe the strength of the relationship. **2**

(ii) Find the equation of the Least -Squares Regression Line. **1**

(i) $r = 0.9592$
(calculator)
$\therefore$ strong correlation positive

(ii) from calculator
$A = -8.2368$
$B = 0.93203$
$y = A + Bx$
$y = -8.2368 + 0.93203x$

Q12 Caringbah

The table below shows the English marks (x) and the Mathematics marks (y) for a class of 12 students (A-L). Only the English mark is available for student L.

	A	B	C	D	E	F	G	H	I	J	K	L
x	67	61	65	67	75	75	69	85	85	89	87	80
y	58	64	66	68	70	72	72	76	80	82	84	

(a) Calculate the correlation coefficient between x and y for the students A to K. Describe the nature of the correlation coefficient between x and y. **2**

(b) Find the equation of the least squares regression line of y on x for the students A to K. Estimate the Mathematics mark of student L. **2**

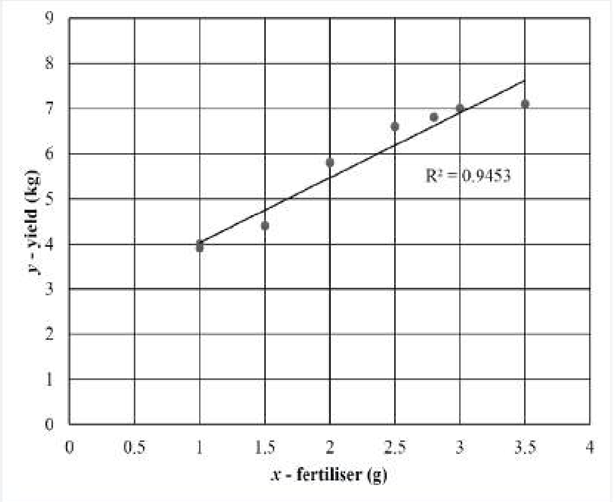
(a) $r = 0.9$, strong positive correlation

(b) $y = 18 + 0.72x$.
when $x = 80$, $y = 18 + 0.72(80) = 75.6$
$\therefore L \doteqdot 76$.

Q13 Blacktown

A biologist assumes that there is a linear relationship between the amount of fertiliser supplied to tomato plants and the subsequent yield of tomatoes obtained. Eight tomato plants, of the same variety, were selected at random and treated, weekly, with a solution in which x grams of fertiliser was dissolved in a fixed quantity of water. The yield, y kilograms, of tomatoes was recorded.

Plant	A	B	C	D	E	F	G	H
x	1.0	1.0	1.5	2.0	2.5	2.8	3.0	3.5
y	3.9	4.0	4.4	5.8	6.6	6.8	7.0	7.1



a) Use the scatterplot to describe the association between 'yield' and 'fertiliser' in terms of strength and direction. **1**

b) Determine the equation of the least-squares regression line for this data. Round your values to two significant figures. **2**

c) A plant with 2.2 grams of fertiliser was not recorded by accident. Calculate the predicted yield for this plant using your answer in part b). **1**

d) Explain why you should not extrapolate from this data to find the yield for high rates of fertiliser usage. **1**

a Linear, strong, positive

b Calculator – stat mode
$y = A + Bx$
$A = 2.5920\ldots$
$B = 1.4371\ldots$
$y = 1.4x + 2.6$

c $y = 1.4 \times 2.2 + 2.6$
$y = 5.68\ kg$
Or $y = 1.5 \times 2.2 + 2.5$
$y = 5.8\ kg$

d There could be a point where fertiliser will not produce extra yield. Reasons could include: high volume of fertiliser may cause damage to the plant; other factors may limit yield, eg sunlight/water; plants may not be able to absorb fertiliser at that rate.

Q14 Ace

The table below shows forearm length and hand length.

Forearm (in cm)	25.0	25.6	26.0	26.6	27.0	27.4	28.0	28.6	29.0	29.2
Hand (in cm)	17.2	17.6	18.2	18.4	19.0	19.0	19.8	19.8	20.4	20.6

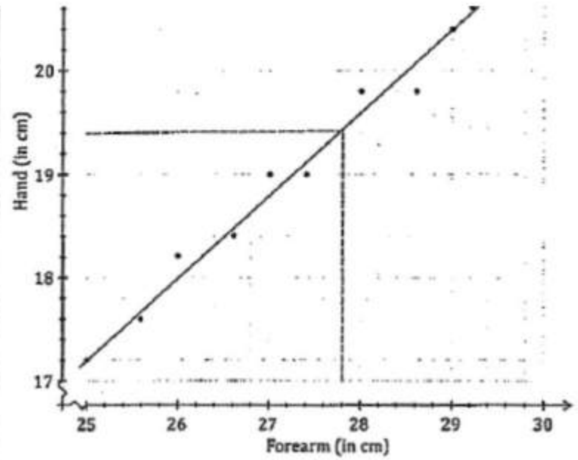
(a) Draw a scatterplot using the above table. **1**

(b) Draw a line of best fit on the scatterplot. **1**

(c) Charlotte has a forearm whose length is 27.8 cm. What is her expected hand length? **1**

(d) Calculate the value of the Pearson's correlation coefficient. Answer correct to four decimal places. **2**

(a)



(b) See line of best fit on the above scatterplot.

(c) When forearm length = 27.8 then hand length = 19.4 cm (from the scatterplot)
∴ Charlotte's hand length should be 19.4 cm.

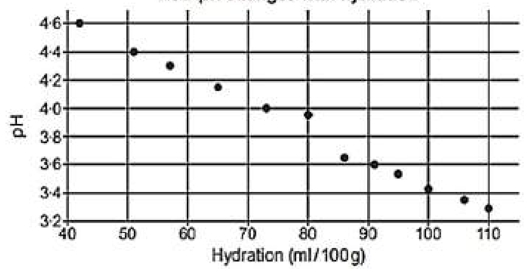
(d) Use the calculator to find Pearson's correlation coefficient.
$r = 0.990691\ldots$
≈ 0.9907

Q15 North Sydney Girls

A baker researches how the pH of sourdough (y) changes with the hydration (x). Hydration is measured in mL of water per 100 g of flour (mL/100g).
The results of his research are shown in the table and diagram below:

Hydration	42	51	57	65	73	80	86	91	95	100	116	110
pH	4.60	4.40	4.10	4.18	4.00	3.98	3.65	3.60	3.55	3.42	3.37	3.10

How pH changes with hydration



(a) Describe the relationship between pH and hydration. **1**

(b) Find the equation of the least-squares regression line using your calculator. Use this equation to estimate the pH of the sourdough when the hydration is 20 mL/100 g. Also comment on the reliability of this estimate. **3**

(a) There is a strong negative linear correlation between pH and hydration.

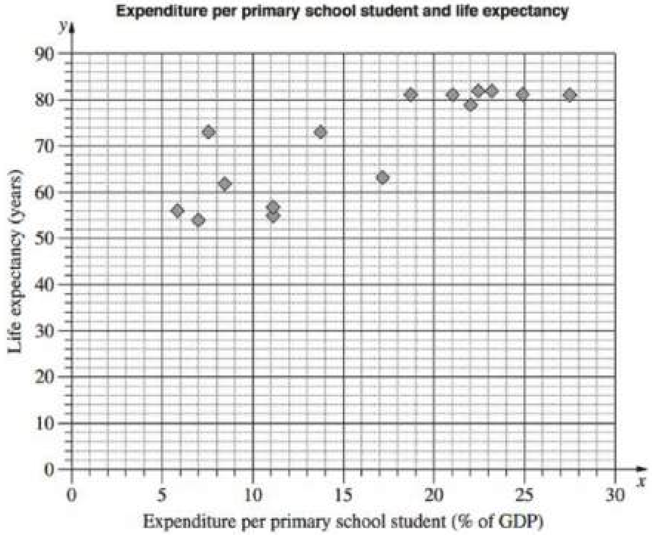
(b) $y = 5.328 - 0.0186x$

$x = 20, \quad y = 5.328 - 0.0186 \times 20$
$= 4.956$

This estimate is not reliable as we are extrapolating outside of the range of data given.

Q16 Hunters Hill

The scatterplot shows the relationship between expenditure per primary school student, as a percentage of a country's Gross Domestic Product (GDP), and the life expectancy in years for 15 countries.



(a) For the data representing expenditure per primary school student, the lower quartile, Q_L is 8.4 and upper quartile, Q_U is 22.5. What is the interquartile range? **1**

(b) Another country has an expenditure per primary school student of 47.6% of its GDP. Would this country be an outlier for this set of data? Justify your answer with calculations. **1**

(c) On the scatterplot, draw the least-squares line of best fit, $y = 1.29x + 49.9$ **2**

(d) Using this line, or otherwise, estimate the life expectancy in a country which has an expenditure per primary school student of 18% of its GDP. **1**

(e) Why is this line NOT useful for predicting life expectancy in a country which has expenditure per primary school student of 60% of its GDP? **1**

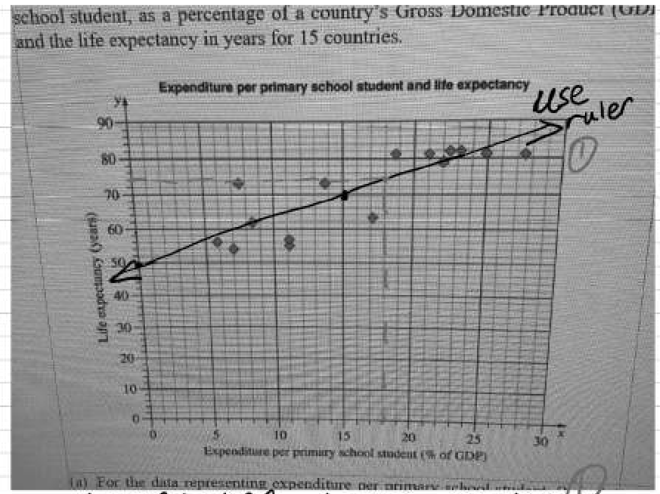
a) $IQR = Q_U - Q_L = 22.5 - 8.4 = 14.1$

b) $Q_U + 1.5 \times IQR = 22.5 + 1.5 \times 14.1 = 43.65\%$

Since $47.6\% > 43.65\%$

∴ It is an outlier

c)



$y = 1.29x + 49.9$ $\quad b = 49.9, \ m = 1.29$ (1)

$mx + b$ $\quad x = 15, \ y = 69.25 \ (15, 69.25)$

d) from the diagram, at $x = 18\%$

$y = 74$ (answ 72-74 all acceptable) (1)

∴ the life expectancy is 74.

e) At 60% GDP, the line predicts a life expectancy over 100 which exceeds the expected life span for most human. So this line of best fit is only predictive in a lower range of GDP expenditure. (1)

Q17 Western

Max did a survey of a group of people he knew about their age and how much they earn each week. The results are shown in the table below.

Age (years) (x)	18	45	28	15	32	68
Wage ($/week) (W)	715	2350	1530	438	1690	1320

(a) Using your calculator find (r) the correlation coefficient and explain what type and strength of correlation this data gives. **2**

(b) Using your calculator find the equation of the least-squares regression line in the form **1**

$W = Bx + A$ where A and B are integers.

(c) Use your equation to estimate the earnings of a 50 year-old worker. **1**

(d) Could your equation from part(b) be used to make valid estimates for ages greater than 68 and less than 15 years? **2**
Validate your response with calculations and or reasons.

a)

Scatterplot does not need to be drawn to answer the question, however can help as a visual aide.
Using calculator $r = 0.5263217513$ $\quad r = 0.53$ (2dp)
This means a weak/medium positive correlation, due to the final data point being included.

b) $W = Bx + A$
From the calculator
$B = 18.47948276$ so $B = 18$ $\quad A = 706.0377586$ so A = 706
$\therefore W = 18x + 706$

c) $W = 18x + 706$ when $x = 50$
$W = 18 \times 50 + 706 = \$1606$

d) The equation is not a valid way to extrapolate for younger and older workers, as it has only a moderate correlation coefficient. There is one data point (for a 68-year old) which affects the results causing this.

People below 15 would not be working because they are too young, and at the older end, people have retired or are working less, and this equation does not take this into account.

The equation is only valid for interpolation between 15 and 45 and, even then, has been affected by the last data point, a possible outlier. Max would need more data for ages outside of this if he is to make a more accurate prediction. Had he left the last data point out the correlation would have been much stronger and more accurate in the 15 – 45yr range.

Q18 CSSA

Seven students, who are still on their Learner's Permit, recorded the number of driving hours they have logged so far and the mark they scored in their Visual Art assessment task.

The results are recorded in the table below:

Name	Ally	Bree	Cain	Dan	Elle	Frank	Guy
Hours of driving (x)	63	68	72	76	82	84	91
Visual Art mark (y)	73	71	71	85	92	80	81

(a) Calculate Pearson's correlation coefficient for the data, correct to two decimal places. **1**

(b) Calculate the value, to two decimal places, of the gradient and the y-intercept. Use these values to write the equation of the least-squares line of best fit. **2**

(c) Another student, who has logged 55 hours of driving and who scored 94 in Visual Art, is added to the data. **1**
Describe the impact this addition to the data will have on Pearson's correlation coefficient.

(d) A student made the statement, "Doing well in Visual Art depends on how much driving you have done." **1**
Based on the data, justify whether you agree or disagree with the statement.

(a) $r = 0.63$

(b) $m = 0.51$
$c = 40.16$
$y = 0.51x + 40.16$

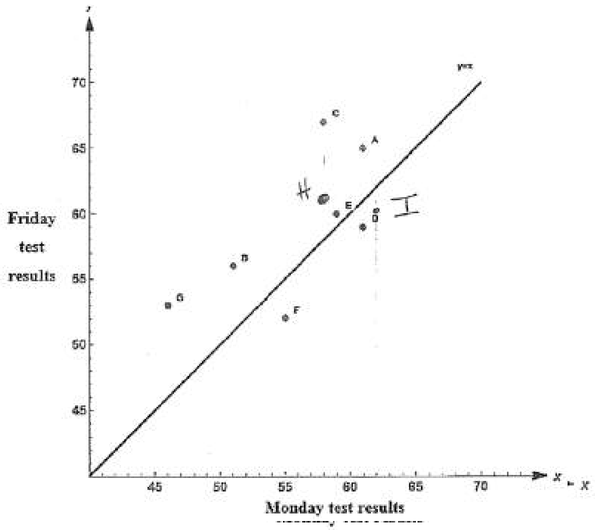
(c) The correlation coefficient will be lower/weaker when the additional student is included in the data. The value is now $r = 0.009$

Q19 Cecil Hills

After a class of ten students sits a test on Monday, the teacher spends time doing revision and tests the students again on Friday. Only nine of the students are present for the second test on Friday. The results, in percentages, for those nine students, are shown in the following table.

Student	A	B	C	D	E	F	G	H	I	J
Monday (M)	61	51	58	61	59	55	46	58	62	63
Friday (F)	65	56	67	59	60	52	53	61	60	Absent

(a) Add the results of students H and I to the scatter plot. **1**



(b) Determine the equation of the least squares-regression line of best fit. **1**

(c) Using your answer to part (b), predict the result for the student who was absent for Friday's test. Give your answer to the nearest whole mark. **1**

(d) The scatterplot above also shows the line $y = x$. Explain the significance of a student being represented below the line $y = x$. **1**

(b) $y = 0.62x + 23.81$

(c) $x = 63$

$y = 0.62 \times 63 + 23.81$

$= 62.87 \approx 63$

(d) student who appears below the line $y = x$ performed better on Monday's test